

ADRITEYO DAS

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EDUCATION

Manipal Institute of Technology, Manipal, Karnataka

2023 – 2027

Bachelor of Technology — Computer and Communication Engineering

CGPA: 9.26

(Top 0.01 percentile in ICT Department)

SKILLS

Software Engineering	Backend Development, Full-Stack Development, Distributed Systems, REST API Development
Data Engineering	ETL/ELT Pipelines, Batch Processing, Data Quality
Machine Learning	Computer Vision, Natural Language Processing (NLP), Explainable AI (XAI), Agentic Workflows
Security	Reverse Engineering, Malware Analysis, Digital Forensics & Incident Response (DFIR), Network Security

TECHNOLOGIES

Languages	Python, Go, C/C++, Java, SQL, Bash
Cloud & DevOps	AWS, Terraform, Docker, Jenkins, Spinnaker
Data Engineering	Apache Spark, PySpark, Databricks, DynamoDB
Observability	Splunk, OpenSearch
AI	PyTorch, TensorFlow, Hugging Face Transformers, Google ADK
Backend	Flask, REST APIs
Databases	MySQL, MariaDB
Developer Tools	Git, Linux, CUDA, Ghidra, IDA Pro

EXPERIENCE

Software Engineer Program (SEP) Intern

June 2026 – August 2026

JPMorgan Chase & Co. — Corporate Technology

Bengaluru, India

- Worked as a Software Engineer on internal data platform infrastructure and developer tooling.
- Built observability and monitoring capabilities for enterprise ETL pipelines.
- Developed custom agentic workflows for AI-assisted diagnostics and operational support.
- Automated internal CI/CD workflows to improve deployment and operational efficiency.
- Contributed across the full stack to build internal engineering tools and user-facing interfaces.

Research Intern

January 2025 – August 2025

Indian Statistical Institute, Kolkata — CVPR Research Group

Kolkata, West Bengal, India

- Collaborated with Dr. Umapada Pal alongside international researchers from University of Salford and Chinese Institute of Information Science.
- Implemented novel **deepfake detection algorithm** utilizing **non-correlative feature interaction** that outperformed state-of-the-art (SOTA) methods.
- **Published research paper accepted at ACPR 2025** (Asian Conference on Pattern Recognition), demonstrating significant advancement in multimedia forensics.
- Applied **advanced computer vision techniques**, **CNNs**, **feature engineering**, and **ensemble learning** for real vs. fake scene classification.

Undergraduate Research Assistant

July 2024 – Present

Manipal Institute of Technology

Manipal, Karnataka, India

- Working with **Dr. Nisha P. Shetty** on explainable AI (XAI), multimodal deep learning, cross-lingual NLP, and medical image analysis.
- **First-author publication in Frontiers in AI (Q1 journal)** on explainable multimodal skin lesion classification achieving **98% accuracy** using **attention-based fusion mechanisms**, **Grad-CAM visualization**, and **Integrated Gradients interpretability**.
- Conducting research on **criminal intent detection in social media text** via **transformer-based models (DistilBERT, Electra fine-tuning)**, **FastText embeddings**, and **hybrid feature extraction** within a **PCC + ACO optimized ensemble framework**; achieved **ROC-AUC: 0.9618** (manuscript under preparation).
- Expertise in **NLP**, **sentiment analysis**, **transfer learning**, **hyperparameter optimization**, and **adversarial robustness**.

Head of Research & Development

April 2024 – Present

Cryptonite — MIT Manipal (Rank #2 India on CTFtime)

Manipal, Karnataka, India

- Leading **AI-cybersecurity research initiatives** for one of Asia's premier academic CTF teams, bridging **machine learning** and **information security**.

- Spearheading projects including **audio profanity detection using deep learning**, **privacy-preserving OSN graph generation**, and **AI-driven threat detection systems**.
- Developed and deployed **forensics and security challenges for niteCTF 2024 and 2025** (3rd highest-rated Indian CTF globally on CTFtime).
- Mentoring junior members on **adversarial machine learning**, **neural network security**, **malware analysis**, **digital forensics (DFIR)**, and **incident response**.
- Active competitor in national/international CTF competitions, specializing in **Digital Forensics**, **Reverse Engineering**, **Binary Exploitation**, and **Network Security**.

ACHIEVEMENTS

- **1st Position:** EnigmaXplore 2024 CTF (IIIT Naya Raipur), BITSCTF 2024 (BITS Goa), CruXipher 2024 (BITS Hyderabad), KashiCTF 2025 (IIT BHU Varanasi).
- **2nd Position:** TexSAW CTF 2025 (UT Dallas), BITSCTF 2025 (BITS Goa), CapturePoint CTF 2025 (IIIT Vadodara), ExploitX CTF 2025 (Top Individual).
- **3rd Position:** Codefest CTF 2025 (IIT BHU), eHax CTF 2025 (DTU Delhi), FooBar CTF 2025 (NIT Durgapur).
- **Achiever's Scholarship** awarded by Manipal Institute of Technology for outstanding academic performance.
- **Finalist** at JPMorgan Code for Good hackathon for social impact software development.

PROJECTS

- **rpsmon: Rust TUI System Resource Monitor** 2026
Rust, TUI, Systems Programming, Process Monitoring, macOS/Linux, Homebrew
 - Built **fast cross-platform terminal system monitor** for macOS and Linux with live CPU, memory, thread, status, and disk I/O telemetry.
 - Implemented **tree navigation**, **watchlists**, **lock-selection mode**, configurable alerts, detail views, filtering, and interactive process signal controls.
 - Published optimized release builds with **LTO/stripping**, crates.io installation support, and a Homebrew formula for developer-friendly distribution.
- **Xerxes: High-Performance Filesystem Navigation Engine** 2026
Rust, redb, nucleo-matcher, notify, Unix Sockets, FZF/TUI, Systems Programming
 - Built **cross-platform directory jumping engine** for macOS and Linux with fuzzy matching and frequency-based ranking.
 - Designed **client-daemon architecture** using Unix domain sockets, background indexing, and real-time filesystem monitoring to minimize cold-start latency.
 - Added **smart learning**, **aliases**, **shell setup automation**, FZF integration, and a robust TUI fallback for zero-configuration developer workflows.
- **ChiefWarden: Hybrid Malware Detection System** 2024
PyTorch, XGBoost, PE Analysis, Feature Engineering, Ensemble Learning
 - Developed **hybrid malware detection system** combining **static PE analysis with machine learning models** achieving **98.89% accuracy**.
 - Implemented **ensemble classifier using XGBoost and neural networks** for robust threat identification and classification.
 - Integrated **feature extraction pipelines** for PE headers, section analysis, import/export tables, and behavioral signatures.
- **SimuTrade: Real-Time Stock Trading Simulator with AI Forecasting** July 2025
React.js, Flask, MySQL, JWT, Prophet, yFinance, REST APIs, Docker, RBAC, Time-Series Forecasting
 - Engineered full-stack **real-time stock trading platform** with **JWT-based authentication**, **role-based access control (RBAC)**, and microservices architecture.
 - Integrated **Prophet time-series forecasting models** for predictive trend analytics using live market data via **yFinance API**.
 - Implemented **Redis caching layer** to optimize API call efficiency and reduce latency by 60%.
 - Developed **portfolio management system**, **PnL computation engine**, and **gamified community features** with leaderboard rankings.
 - Built comprehensive **financial literacy learning center** with curated educational content and interactive modules.
 - Containerized application using **Docker** for seamless deployment; documented RESTful API endpoints via Postman.
- **DropLAN: Peer-to-Peer Local Network Sharing Platform** July 2025
Flask, WebSockets, Socket.IO, React.js/Vite, Docker, SHA-256, QR Code Generation, LocalStorage
 - Built full-stack **LAN-based file and note sharing system** with **real-time bidirectional data transfer via WebSockets**.
 - Implemented **temporary bucket system with 30-minute TTL expiry**, secure file hashing (SHA-256), and **live synchronization using Socket.IO**.
 - Developed **QR code-based portal access**, automatic IP detection, and persistent notepad with **localStorage autosave functionality**.
 - Created custom CLI tool DropLAN for one-click deployment; designed minimalist UI with smooth animations and monospace typography.

- **MemHawk: AI-Powered Memory Forensics Platform** 2024
Python, Ollama, Electron, Volatility, Digital Forensics, LLM Integration
 - Created **AI-enhanced memory forensics platform** integrating **Volatility Framework with large language models (Ollama)**.
 - Developed **Electron-based desktop application** for comprehensive memory dump analysis with natural language querying capabilities.
 - Automated **incident response workflows** and artifact extraction for digital forensics investigations.
- **Astraeus: Orbital Collision Risk Prediction System** 2024
Three.js, XGBoost, Flask, Machine Learning, WebGL, Real-Time Visualization
 - Built **real-time satellite collision risk prediction system** using **ensemble machine learning (XGBoost)** achieving **94.2% accuracy**.
 - Developed interactive **3D orbital visualization using Three.js and WebGL** for space traffic monitoring.
 - Integrated predictive analytics with Flask backend for real-time risk assessment and alerts.

OPEN SOURCE CONTRIBUTIONS

- **Google Mandiant macOS UnifiedLogs** — Contributed to the macOS Unified Logs parsing toolkit used in DFIR workflows. [PR #97](#)
- **CERN CppInterOp** — Improved C++ interoperability infrastructure for Cling/LLVM-based interactive compilation workflows. [PR #842](#)
- **Z3 Solver** — Contributed to Microsoft's open-source SMT theorem prover, strengthening solver tooling and reliability. [PR #9500](#)

PUBLICATIONS

- **Das, A., et al. (2025). Non-correlative Feature Interaction for Classification of Real and Fake Scene Images.** *8th Asian Conference on Pattern Recognition (ACPR)*, Gold Coast, Australia. Springer. [DOI: 10.1007/978-981-95-4398-4_17](#)
- **Das, A., Shetty, N. P., et al. (2025). Comparative Analysis of Multimodal Architectures for Effective Skin Lesion Detection.** *Frontiers in Artificial Intelligence*, 8, 1608837. [DOI: 10.3389/frai.2025.1608837](#)
- **Das, A., Shetty, N. P., et al. (2026). Enhanced Stress Detection via Intermediate Fusion of Intermodal Signals.** *Intelligence-Based Medicine*, 15, 100406. Elsevier. [DOI: 10.1016/j.ibmed.2026.100406](#)

LEADERSHIP & VOLUNTEERING

- **Head of Deep Learning** — The Data Alchemists (Official Data Science Club, MIT Manipal)
- **Head of Cybersecurity** — Research Society Manipal, MIT Manipal
- **Active Member** — Rotaract Club of Manipal (Community Service)

This resume was last updated on 2026-07-28.